

Institute of Mathematical Sciences
University of Malaya
SIM1001 Basic Mathematics
SIX1013 Fundamentals of Advanced Mathematics

TUTORIAL 1 (Part II)

1. For each of the following matrices, determine whether the matrix is row-echelon form, reduced row-echelon form, both, or neither.

(a) $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 0 & 1 & 2 & 0 & 8 & 2 \\ 0 & 0 & 1 & 7 & 3 & 4 \\ 0 & 0 & 1 & 2 & 0 & 8 \end{bmatrix}$

(f) $\begin{bmatrix} 0 & 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 & 0 & 8 \end{bmatrix}$

(g) $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

(h) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

(i) $\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$

(j) $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$

2. Show that the matrix A below is row equivalent to a reduced row-echelon matrix.

$$A = \begin{bmatrix} 0 & -3 & 2 & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & -3 \\ 0 & 1 & 2 & -1 & 0 & 1 \\ 0 & 0 & -4 & 0 & -2 & 4 \end{bmatrix}$$

3. For each of the following systems of linear equations, write the augmented matrix of the system and reduce the matrix to row-echelon form. Hence, determine whether each of the following system is consistent or inconsistent. Solve the linear system if it is consistent.

(a) $\begin{aligned} x + y &= 1 \\ 2x + 2y &= -1 \end{aligned}$

(b) $\begin{aligned} x - 2y + z &= -1 \\ y - 3z &= 5 \\ z &= 2 \end{aligned}$

(c) $\begin{aligned} x - 2y + z &= -1 \\ 2x - 3y - z &= 3 \\ x - 2y + 2z &= 1 \end{aligned}$

(d) $\begin{aligned} x + 2y - 2z &= 1 \\ -x + z + w &= -2 \\ 3y - 3z + 3w &= 1 \end{aligned}$

(e) $\begin{aligned} 2x - y + z &= 1 \\ -x + z &= -2 \\ 3x + 2y - 2z &= -1 \\ x - 2y + 3z &= -2 \\ 5x + 2y + 6z &= -1 \end{aligned}$

4. Solve each of the following systems of linear equations by Gaussian elimination.

$\begin{aligned} x + y + z &= 4 \\ \text{(a)} \quad -x - y + z &= -2 \\ 2x - y + 2z &= 2 \end{aligned}$	$\begin{aligned} x + 2y - 2z &= -1 \\ \text{(b)} \quad 3x - y + 2z &= 7 \\ 5x + 3y - 4z &= 2 \end{aligned}$
$\begin{aligned} 4x - 8y - 3z + 2w &= 13 \\ \text{(c)} \quad 3x - 4y - z - 3w &= 5 \\ 2x - 4y - 2z + 2w &= 6 \end{aligned}$	$\begin{aligned} x - y + 2z &= 5 \\ \text{(d)} \quad 2x + y &= 2 \\ x + 8y - z &= 3 \\ -x - 5y - 12z &= 4 \end{aligned}$

5. Solve each of the following systems of linear equations by Gauss-Jordan elimination.

$\begin{aligned} x - 6y - 4z &= -5 \\ \text{(a)} \quad 2x - 10y - 9z &= -4 \\ -x + 6y + 5z &= 3 \end{aligned}$	$\begin{aligned} x + y + z &= 4 \\ \text{(b)} \quad 3x - y - z &= 2 \\ x + 3y + 3z &= 8 \end{aligned}$	$\begin{aligned} 3x - y + z + 2w &= -2 \\ \text{(c)} \quad x + 2y - z + w &= 1 \\ -x - 3y + 2z - 4w &= -6 \end{aligned}$
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(d) The question 4 (d)

6. Consider the system of linear equations

$$\begin{aligned} 3x + 4y - 5z &= -8 \\ x - 2y + z &= 2 \end{aligned} \quad \text{(I)}$$

- (a) Verify that $x = 1, y = 1$ and $z = 3$ is a solution of the system (I).
 (b) Write down the general solution of the system (I).
 (c) Using the obtained general solution, write down two other solutions to linear system (I).

7. Determine the values of a and b so that the linear system

$$\begin{aligned} ax + bz &= -2 \\ ax + ay + 4z &= 4 \\ ax + 2z &= b \end{aligned}$$

- (a) is inconsistent;
 (b) has exactly one solution;
 (c) has infinitely many solution and the general solution has exactly one arbitrary parameter;
 (d) has infinitely many solution and the general solution has exactly two arbitrary parameters.

8. Determine the value of k such that the following linear system

$$\begin{aligned} x + y + z &= 1 \\ 2x + ky + 2z &= 2 \\ 4x + 4y + k^2z &= 2k \end{aligned}$$

- has (a) has no solution; (b) unique solution; (c) infinitely many solution.